

Diabetes Risk Prediction — Final Report

Automated Modeling & Explainability Summary

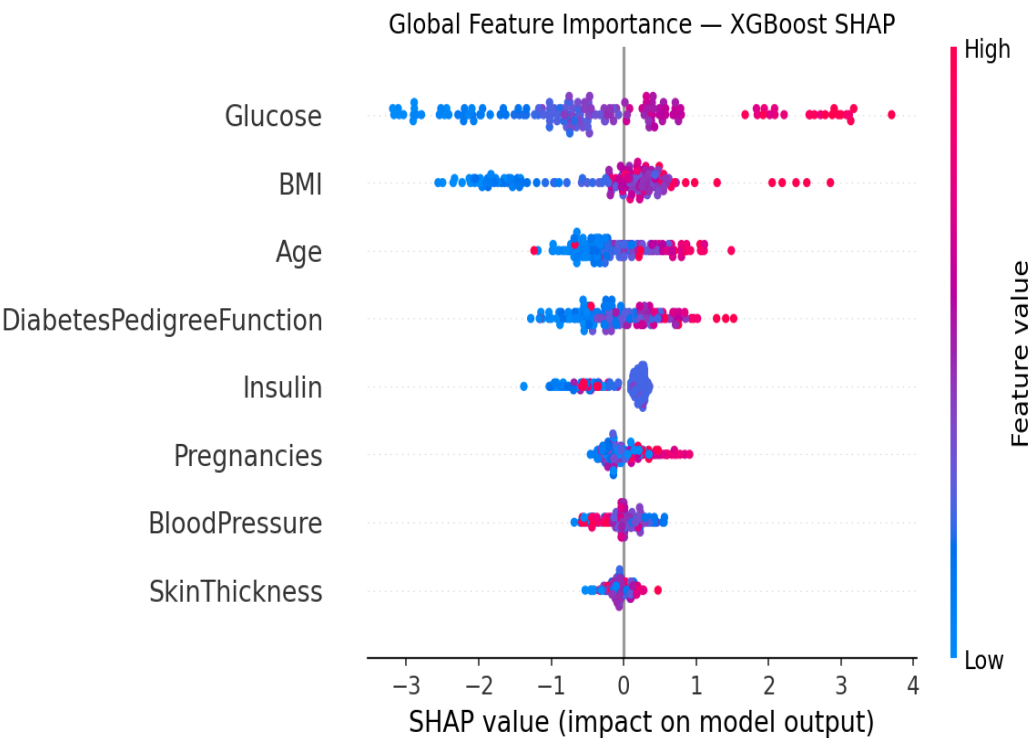
Generated: 2026-04-18 03:03

This report summarises data preprocessing, baseline model training, evaluation, and SHAP-based explainability for the Diabetes Risk Prediction pipeline. Phase 2 fixes the SHAP explainer to use XGBoost TreeExplainer with correct base_values (~0.35).

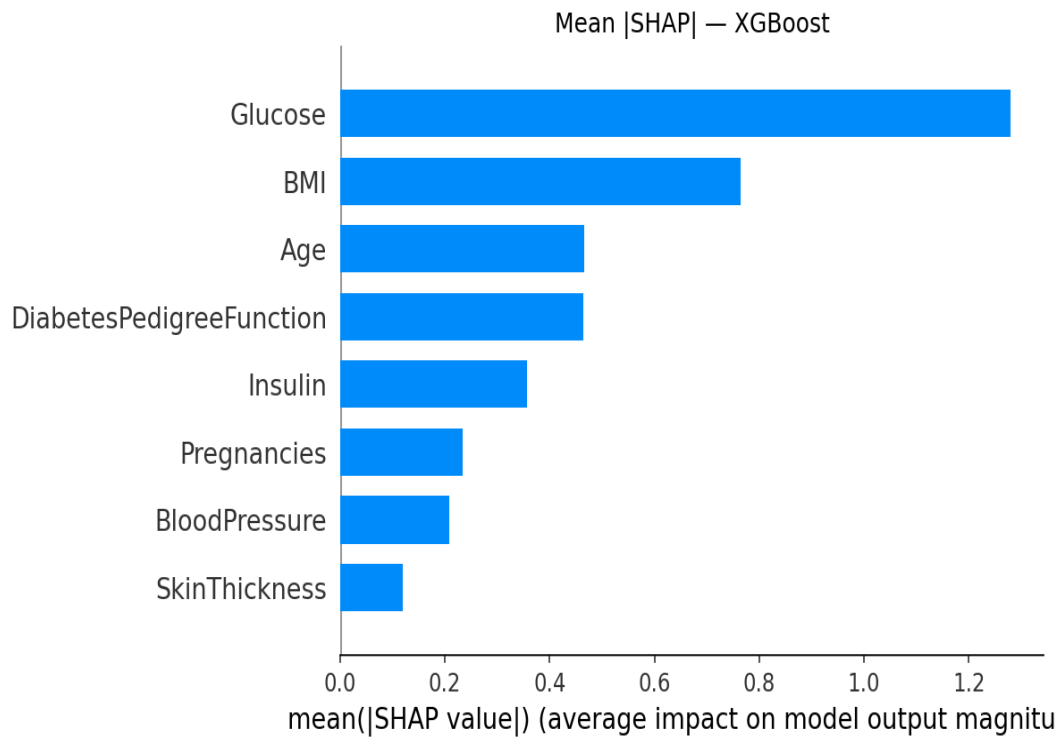
Model Performance Summary

Model	Accuracy	Precision	Recall	F1	ROC_AUC	PR_AUC
LogisticRegression	0.7013	0.5870	0.5000	0.5400	0.8128	0.6732
RandomForest	0.7403	0.6522	0.5556	0.6000	0.8173	0.6971
XGBoost	0.7597	0.6604	0.6481	0.6542	0.8243	0.6826
LightGBM	0.7273	0.6200	0.5741	0.5962	0.8172	0.6701

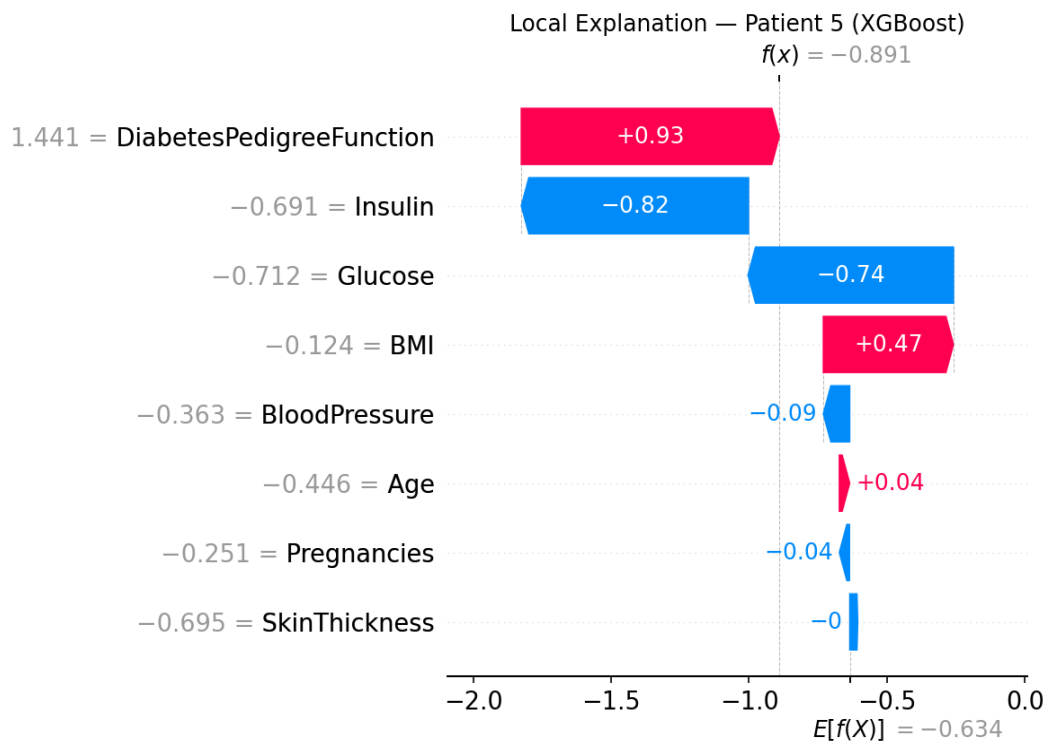
SHAP Summary Plot — XGBoost (Phase 2)



Mean |SHAP| Feature Importance



Local Explanation — Patient 5



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